Feature-Selection High-Resolution Network With Hypersphere Embedding for Semantic Segmentation of VHR Remote Sensing Images

Hanwen Xu, Xinming Tang, Bo Ai, Fanlin Yang, Zhen Wen, and Xiaomeng Yang

***Abstract*— Very-high-resolution (VHR) remote sensing images contain various multiscale objects, such as large-scale buildings and small-scale cars. However, these multiscale objects cannot be considered simultaneously in the widely used backbones with a large downsampling factor (e.g., VGG-like and ResNet-like), resulting in the appearance of various context aggregation approaches, such as fusing low-level features and attention-based modules. To alleviate this problem caused by backbones with a large downsampling factor, we propose a feature-selection high-resolution network (FSHRNet) based on an observation: if the features maintain high resolution throughout the network, a high precision segmentation result can be obtained by only using a 1 × 1 convolution layer with no need for complex context aggregation modules. Specifically, the backbone of FSHRNet is a multi-branch structure similar to HRNet where the high-resolution branch is the principal line. Then, a lightweight dynamic weight module, named the feature-selection convolution (FSConv) layer, is presented to fuse multiresolution features, allowing adaptively feature selection based on the characteristic of objects. Finally, a specially designed 1 × 1 convolution layer derived from hypersphere embedding is used to produce the segmentation result. Experiments with other widely used methods show that the proposed FSHRNet obtains competitive performance on the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the iSAID dataset.**

***Index Terms*— Feature-selection convolution (FSConv) layer, high-resolution backbone, hypersphere embedding, multiscale object.**

I. Introduction

HOW to accurately recognize and segment multiscale objects in very-high-resolution (VHR) remote sensing images is a basic but important research task. Recently, large amounts of semantic segmentation models were proposed with the rapid development of deep learning. Most of them have two important parts: the backbone network and the context aggregation module, where the backbone network extracts features from images, and the context aggregation module enhances the representational ability of features. However, the widely used backbones with large downsampling factor (16× or 32×), e.g., VGG [1], Inception [2], and ResNet [3], are inherently unfriendly to small-scale objects and edges, leading to the requirement for an elaborately designed context aggregation modules, e.g., fusing low-level features [4], multiscale fusion modules [5]–[7], and self-attention-based modules [8]–[10]. Hence, the purpose of this research is to establish a simple but high-precision segmentation model to alleviate the problem caused by backbones with a large downsampling factor.

Nowadays, VGG-like [1], Inception-like [2], [11], [12], and ResNet-like [3], [13]–[15] backbones occupy the main part

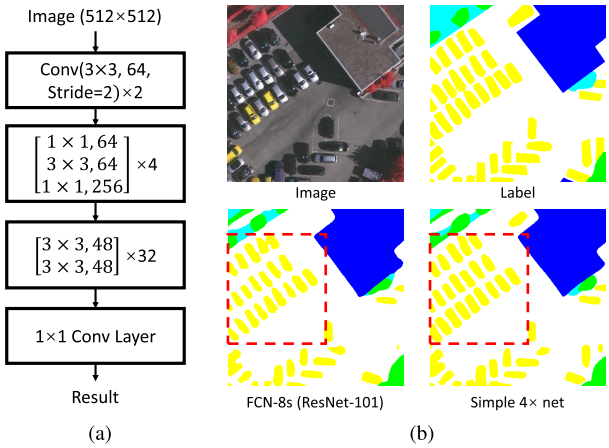


Fig. 1. Advantage of high-resolution features. A simple 4× net is designed to test the effect of high-resolution features by removing some downsampling layers and reducing the number of channels of ResNet-101. Its architecture is depicted in (a) where symbol [ ] represents a residual block [3]. First, this network performs 4× downsampling on the image through two convolution layers. Then, four bottleneck residual blocks with 64 channels and 32 basic residual blocks with 48 channels are used to extract features. Finally, a 1 × 1 convolution layer is added to the end to output the segmentation result without any other context aggregation modules. The comparison experiment in (b) indicates that the simple network with high-resolution features can outperform ResNet-101-based FCN-8s in the segmentation of small objects and edges. The reason for choosing the FCN architecture is that it has no complex context aggregation modules to ensure a fair comparison. More information is illustrated in Table I. (a) Simple 4× Net. (b) Segmentation results.

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of remote sensing semantic segmentation models. The large downsampling factor enables them to learn deep and high-level features but leads to an adverse effect on the segmentation of small-scale objects and edges. Intuitively, maintaining the high-resolution state of features throughout the network may be a direct solution to address this problem. To verify this idea, we conduct a simple experiment and find that, when maintaining high-resolution features (4×) throughout the whole network, high-precision segmentation of small-scale objects and edges can be implemented by simply stacking convolutional layers with a small number of channels, as described in Fig. 1. Table I indicates that the simple 4× network can achieve a competitive precision comparable to ResNet-101-based FCN-8s on ISPRS Vaihingen and Potsdam datasets, but it performs poorly on the iSAID dataset with the imbalanced class distribution. Experientially, this problem can be solved by fusing low-resolution and deep features with high-resolution features. Hence, we design an aligned high-resolution network (Aligned HRNet) by simply modifying the HRNet [16], where each branch has different resolutions, and the high-resolution branch is the principal line, as shown in Fig. 2.

Context aggregation modules are designed to enhance the representational ability of features produced by the backbone. In summary, without adding other auxiliary information (e.g., DSM), the context aggregation modules mainly include: 1) gradually fusing low-level features, e.g., U-shape like structures [4], [17], [18]; 2) receptive field growth and multiscale context fusion, e.g., dilated convolution [19] and multiscale fusion modules [5]–[7]; 3) global context, e.g., self-attention-based methods [8]–[10] and graph-based methods [20]–[23]; and 4) hybrid methods, consisting of two or more different modules. Most of the above modules are designed for the backbones with a large downsampling factor. However, since the HRNet already contains multiresolution features and multiscale receptive fields, there is no need for the U-shape like or enlarging receptive field modules. In order to make better use of multiresolution features, we try to design a hierarchical multiresolution feature fusion module, such as a feature pyramid network (FPN) [24], which allows pixels located on the edges or small objects to assign more weight to the high-resolution features and pixels on large objects to focus on the low-resolution features with large receptive fields. Hence, we present a lightweight feature-selection convolution (FSConv) layer that uses a convolution layer to generate the dynamic convolution kernel to implement adaptively feature selection based on the characteristic of objects. The FSConv increases the segmentation accuracy of multiscale objects and enhances the interpretability of multiresolution features with almost no increase in the calculation.

Unlike the segmentation models with VGG-like, Inception-like, and ResNet-like backbones, our feature-selection HRNet (FSHRNet) directly generates high-resolution fusion features through the Aligned HRNet backbone and FSConv layer. The high-resolution fusion features exhibit good linear separability that achieves high-precision segmentation by using a single 1×1 convolution layer, avoiding the design of a complex context aggregation module. In addition, through borrowing the hypersphere embedding theory from the field of metric learning and face recognition, we analyze the effect of b, x, and W of the 1×1 convolution layer ($y = W^T x + b$) on training and inference, and further improve the segmentation accuracy in some intractable positions. In summary, our main contributions can be summarized as follows.

1) The importance of high-resolution features to VHR images is verified by a simple 4× network, which provides a certain basis for the application of HRNets on VHR images.
2) An FSConv layer is proposed to generate high-resolution fusion features, allowing adaptively feature selection based on the characteristic of objects. The high-resolution fusion features have good linear separability, achieving high-precision segmentation with a single 1×1 convolution layer. The FSConv layer also realizes the decoupling of different resolutions, enhancing the interpretability of multiresolution features.
3) According to the hypersphere embedding theory, we design three variants and ten strategies of the 1×1 convolution layer and draw two conclusions. First, removing bias b when inference can improve the segmentation accuracy on some intractable positions, such as edges. Second, the existence of $\|W\|$ and $\|x\|$ can enlarge the angle between class axes to facilitate the optimization of the neural network.

II. Related Works

A. Backbone Network

In this article, the backbone network represents the feature extraction part of a segmentation model. Regardless of the lightweight backbones [25], [26] on the mobile device and the emerging transformer-like backbones [27], [28], VGG [1], Inception (GoogLeNet) [2], [11], [12], ResNet [3], [13]–[15], EfficientNet [29], and their variants are four types of backbone networks commonly used in semantic segmentation models. The main idea of VGG is to replace the large convolution kernel with multiple small (3×3) convolution kernels to build a deeper network. Early segmentation models mostly used VGG as the backbone, such as SegNet [17], DeconvNet [30], and DeeplabV1 [31]. Inception-like backbones proposed sparsely connected architectures to extract dense and compressed features. At present, a few mainstream segmentation models use Inception as backbone, such as DeepLabV3+ [32] and AERFC [33]. ResNet and its variants are currently the most widely used backbones, which alleviates the overfitting and gradient explosion/disappearance problems through residual learning that occurs when increasing the network depth. EfficientNet [29] consists of the compound coefficient, which studied model scaling and adjusted the depth, width, and the resolution of the network for better performance. Eff-UNet [34] and EfficientPS [35] have obtained high segmentation precision based on EfficientNet. In general, the above backbones follow a common rule: gradually reduce the spatial size of the feature maps, connect the convolution layers from high resolution to low resolution in cascade, and produce a low-resolution representation with a large downsampling factor (16× or 32×). However, this rule is not conducive to the extraction of

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TABLE I

COMPARISON BETWEEN THE SIMPLE 4× NET AND FCN-8S (RESNET-101) IN TERMS OF COMPUTATION COMPLEXITY (GFLOPS, PARAMETERS, AND INFERENCE MEMORY) AND SEGMENTATION ACCURACY (MEAN F1 SCORE, OA, AND MEAN INTERSECTION OVER UNION) ON THE ISPRS VAIHINGEN DATASET, THE ISPRS POTSDAM DATASET, AND THE iSAID DATASET. THE BATCH SIZE IS SET TO 1 WHEN CALCULATING GFLOPS, PARAMETERS, AND INFERENCE MEMORY

Dataset	Img size	Method	GFLOPs	#Params(MB)	Infer Memory(MB)	Mean F1	OA(%)	mIoU(%)
Vaihingen	512×512	FCN-8s (ResNet-101)	52.33	42.52	695.55	84.69	87.60	74.10
		Simple 4× Net	**29.20**	**1.78**	**645.08**	**85.22**	**87.62**	**74.82**
Potsdam	512×512	FCN-8s (ResNet-101)	52.33	42.52	695.55	**89.54**	**89.09**	**81.30**
		Simple 4× Net	**29.20**	**1.78**	**645.08**	89.15	88.46	80.68
iSAID	800×800	FCN-8s (ResNet-101)	127.39	42.52	1749.03	-	-	**57.85**
		Simple 4× Net	**71.21**	**1.78**	**1610.36**	-	-	42.92

edges and small objects in VHR images, as shown in Fig. 1(b). Some segmentation models directly reduce the downsampling factor to 8 to improve the accuracy, such as DeepLabV2 [5] and PSPNet [7].

Maintaining high resolution (4×) is the solution adopted in this article to alleviate the problem caused by the large downsampling factor. Scholars have conducted some studies on high-resolution feature representation. Pohlen *et al.* [36] proposed a two streams network, where one stream stays at the full resolution and the other undergoes a sequence of pooling operations. However, the full-resolution stream just incorporates the low-resolution features instead of performing the convolution layer itself. GridNet [37], MSDNet [38], and HRNet [16] are all multibranch-like networks. GridNet generalized many architectures, including conv-deconv, residual, and U-Net, but increased the redundancy and complexity. MSDNet cannot learn strong high-resolution features as there is no information received from low-resolution features. HRNet is closely related to our backbone network. It presented a repeated multiresolution fusion module to exchange the information across different resolution branches and directly outputted the multiresolution features. Zhang *et al.* [39] proposed a multiscale context aggregation network based on HRNet to aggregate multilevel spatial correlations on remote sensing images.

B. Context Aggregation Module

Regardless of adding other auxiliary information (e.g., DSM), we divide the context aggregation modules of segmentation model into four categories.

1) *Gradually Fusing Low-Level Features:* U-Net [4] and SegNet [17] recovered high-resolution features from low resolution by fusing low-level features and stacking upsampling operations. RefineNet [40] and TreeUNet [18] proposed a multipath refinement network and a Tree-CNN block, respectively, to explicitly exploits the low-level features. Zhou *et al.* [41], Mboga *et al.* [42], and Ji *et al.* [43] used U-shape like network to extract object in remote sensing images.
2) *Receptive Field Growth and Multiscale Context Fusion:* Receptive field represents the ability to obtain context information. To enlarge the receptive field, dilated convolution was introduced in [19], DeepLabV1 [31], and [44]. Later, the multiscale fusion modules were widely studied. DeepLabV2 [5], V3 [6], and PSPNet [7] fused the pyramid features by atrous spatial pyramid pooling (ASPP) and pyramid pool module (PPM), respectively. LKPP [45] and AERFC-SPP [33] further developed the pyramid pooling structure. Ding *et al.* [46] proposed a two-stage multiscale architecture to enlarge the valid receptive field.
3) *Global Context:* Global context modules expand the receptive field to the whole image. Self-attention generates dynamic weights by calculating the similarity between feature vectors, whose various variants have been proposed in the semantic segmentation field, such as DANet [8], CCNet [9], OCR [10], and others for remote sensing images [47]–[50]. Some transformer-based segmentation methods [51], [52] were proposed recently with the rapid development of transformer. Graph-based methods [20]–[23] are another way to realize global context.
4) *Hybrid Methods:* Many hybrid methods [53]–[55] were proposed to increase segmentation accuracy of remote sensing images by integrating multiple modules.

C. Hypersphere Embedding

Because 1×1 convolution layer does not involve spatial information, it can be regarded as performing a fully connected operation ($y = W^T x + b$) for each feature point. According to the hypersphere embedding, $y = W^T x + b$ can be rewritten as $y = \|W^T\|\|x\|\cos(\theta) + b$ by the cosine law. If zeroing the bias b and normalizing W and x, the features can be constrained to a hypersphere manifold. The hypersphere manifold using angle as the measure method has achieved great success in the field of face recognition. SphereFace [56] focused on optimizing the L2-norm of x and angle between x and W by removing the b, normalizing the w and adding a multiplicative angle margin. AM-Softmax [57] normalized the x and expanded the radius of the hypersphere by multiplying a scale factor. CosFace [58] and ArcFace [59] normalized W and x at the same time and replaced the multiplicative margin with an additive margin to reduce the difficulty of optimization.

III. METHODOLOGY

Our goal is to segment multiscale objects in VHR images. The FSHRNet is presented to achieve this goal, which consists of three parts: 1) an Aligned HRNet to produce the multiresolution features; 2) an FSConv to fuse multiresolution features; and 3) a 1×1 convolution layer derived from

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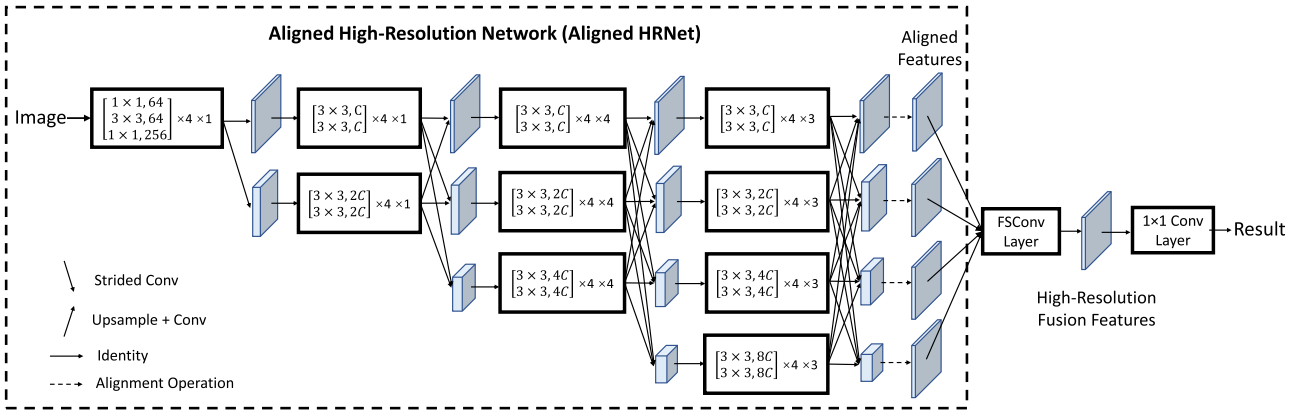


Fig. 2. Network architecture of FSHRNet, including an Aligned HRNet, an FSConv laver, and a 1 × 1 convolution layer. The default number of branches is 4, [ ] represents a residual block, and C is set to 48. The multiresolution features outputted by the high-resolution backbone are aligned to 128 by the 1 × 1 convolution layer and bilinear upsampling to meet the requirement of the FSConv layer.

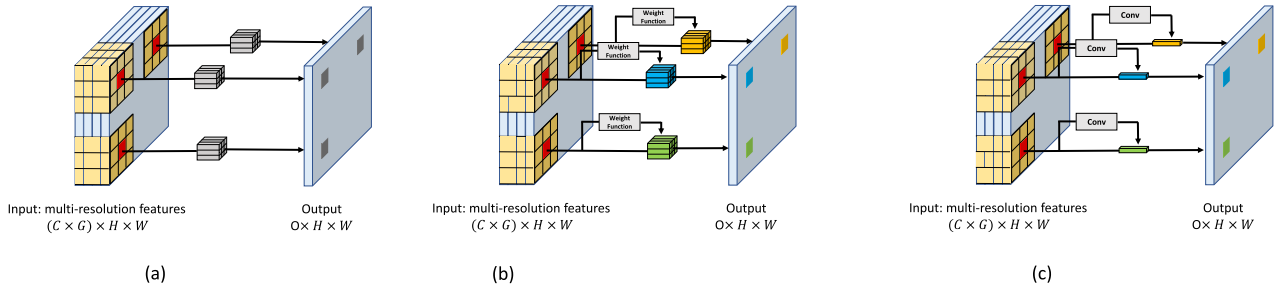


Fig. 3. Illustration of FSConv layer. (a) Standard convolution layer, where each position shares the same and fixed convolution kernel. (b) Proposed framework to realize the dynamic weight for feature selection (framework of feature-selection dynamic weight). The main idea is to design a weight generation function to independently generate a dynamic weight for each position. (c) FSConv layer implemented based on the framework given in (b).

hypersphere embedding to produce the segmentation results. Based on the simplicity principle, the FSConv and hypersphere embedding aim to improve accuracy with minimal parameters and complexity.

A. High-Resolution Backbone Network

Fig. 1 and Table I have shown the advantage of high-resolution features in segmenting small objects. However, due to the insufficient receptive fields and inability to extract deeper features of high-resolution features, trivial patches and holes often appear in some indistinct and large objects, as shown in Fig. 9(b) and (c). This problem is relatively rare in backbones with large downsampling factors, such as ResNet-101, where more context can be considered when classifying a feature point. In pursuit of a larger receptive field and deeper features, a multibranch structure in HRNet is introduced to allow high-resolution features to be fused with low-resolution features to achieve segmentation of multiscale objects.

The high-resolution backbone takes high-resolution convolution as the main stream, gradually adding low-resolution streams with two subsampling factors one by one, as provided in Fig. 2. Each branch has multiple residual blocks, each of which consists of 1 × 1 or 3 × 3 convolution layers. The channels of the convolution kernel are increased from high to low resolutions to obtain deeper features without significantly reducing efficiency. In addition, the repeated multiresolution fusion module in HRNet is introduced to exchange the information across different resolution branches and repeats every four residual blocks. Finally, the 1 × 1 convolution layer and bilinear upsampling are used to align the channel dimensions and spatial size of the multiresolution features. The final alignment operation is the main difference from HRNet, which balances the importance of different resolution features. For convenience, we named it Aligned HRNet in this research.

B. Feature-Selection Convolution Layer

Concatenate and addition operations followed by convolution layer are two widely used fusion methods. The HRNetV2 uses concatenate + convolution layer to fuse multiresolution features. However, due to the local connection and weight sharing of the convolution layer that all positions in feature maps share the same and fixed convolution kernel, adaptively selecting features based on the characteristic of objects cannot be achieved. FPNs [24] in object detection models use multiresolution feature maps to hierarchically detect objects with different scales. Correspondingly, we think that the segmentation of multiscale objects can also benefit from the hierarchical structure; for instance, the pixels located on the edges or small objects should assign more weight to the high-resolution features, and the pixels on large objects should focus on the

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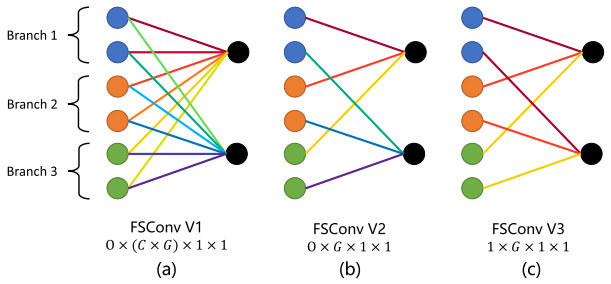


Fig. 4. Comparison of the three version FSConv layers in the channel direction. (a) FSConv V1. (b) FSConv V2. (c) FSConv V3. The left points in each subfigure represent a feature vector taken from any position on the multiresolution feature maps. The right points represent the feature vector produced by FSConv layers. We set $C = 2$, $G = 3$, and $O = 2$ for visualization purposes. Points from the same branch of Aligned HRNet have the same color. The lines between input and output represent the weights of $\varphi^{i,j}$, and the same weight has the same color.

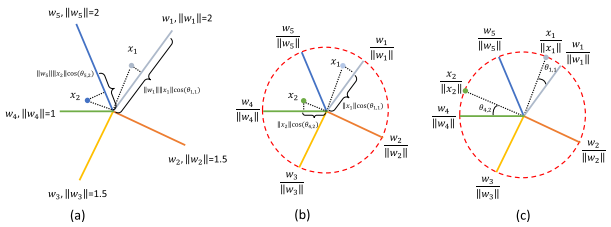


Fig. 5. Geometric interpretation of the three variants of $y = W^T x + b$. (a) Removing bias b. (b) Removing bias b and normalizing W. (c) Removing bias b, normalizing W, and feature vector x. Three variants represent three mappings from feature vectors x to classes. $\|w_j\|$ plays an important role in variant (a) so that x_2 is classified as category 5 rather than category 4. Variant (b) maps W to the unit hypersphere where the classification is only determined by the projection from x_i to w_j. Variant (c) simultaneously maps x and W to the unit hypersphere, in which the angle between x_i and w_j becomes the measure method instead of the Euclidean distance.

low-resolution features with large receptive fields. Hence, our goal is to design a feature-selection dynamic weight method to fuse the aligned multiresolution features generated by the Aligned HRNet, allowing the network to automatically adjust the weights of different resolutions for each position.

We assume that the Aligned HRNet has four branches, and let $X = [X_a, X_b, X_c, X_d] \in R^{(C\times G)\times H\times W}$ denote the aligned multiresolution feature maps, where H and W represent the height and width, C is the channels of a branch, and G is the number of branches. Fig. 3(b) gives the framework of the feature-selection dynamic weight method. The main idea is to design a weight generation function to independently generate dynamic weight φ for each position. There are two problems to solve: 1) the feature aggregation scope of each dynamic weight and 2) how to design the weight generation function. On the first question, since the goal is to assign weights of different resolution features to each position, the feature aggregation scope can be limited to the channel direction of a position without considering the spatial scope. For simplicity, we model the dynamic weight as a 1×1 convolution kernel. On the second question, borrowing from deformable convolution [60], the convolution layer is used as the weight generation function. In summary, we use the convolution layer to generate a dynamic 1×1 convolution kernel for each position, as shown in Fig. 3(c). We named it the FSConv layer. Its generalized form is given as follows:

$$F_{o,i,j} = \sum_{c=1}^{C\times G} \varphi_{o,c}^{i,j} X_{c,i,j} \tag{1}$$

$$\varphi^{i,j} = \text{Conv}(X_{K_{i,j}}) = W X_{K_{i,j}} \tag{2}$$

where $\varphi^{i,j}$ is the generated 1×1 convolution kernel for position (i, j), $K_{i,j}$ indexes the set of pixels that $\varphi^{i,j}$ is conditioned on, W represents the convolution kernel to generate φ, and $F \in R^{O\times H\times W}$ is the produced fusion features.

Specifically, we design three versions of FSConv from complex to simple by changing the feature aggregation scope of $\varphi^{i,j}$. FSConv V1 models $\varphi^{i,j}$ as a standard 1×1 convolution kernel, that is, $\varphi^{i,j} \in R^{O\times(C\times G)\times1\times1}$ and $W \in R^{(O\times(C\times G)\times1\times1)\times(C\times G)\times K\times K)}$. However, generating a complete convolution kernel φ for each pixel consumes too much memory. Hence, we simplify the $\varphi^{i,j} \in R^{O\times(C\times G)\times1\times1}$ to $\varphi^{i,j} \in R^{O\times G\times1\times1}$ by sparse connection where each output channel is only weighted by its corresponding channel at different resolution branches, such as SKNet [61]. We named it FSConv V2 and implement it by modifying the convolution kernel W and adding a softmax function, as described in the following formula:

$$F_{o,i,j} = \sum_{g=1}^{G} \varphi_{o,g}^{i,j} X_{(g-1)\times C+o,i,j} \tag{3}$$

$$\varphi^{i,j} = \text{Softmax}(\text{Conv}(X_{K_{i,j}})). \tag{4}$$

Furthermore, we consider it unnecessary to have different weights on the same resolution branch. Then, we simplify $\varphi^{i,j} \in R^{O\times G\times1\times1}$ to $\varphi^{i,j} \in R^{1\times G\times1\times1}$ by sharing weight at features from the same branch and name it FSConv V3. Group convolution and softmax function are used to implement it as described in the following formula:

$$F_{o,i,j} = \sum_{g=1}^{G} \varphi_{g}^{i,j} X_{(g-1)\times C+o,i,j}$$

$$\varphi^{i,j} = \text{Softmax}(\text{GroupConv}(X_{K_{i,j}})). \tag{5}$$

The key to distinguish the three versions of FSConv is the different aggregation scope of $\varphi^{i,j}$ in the channel direction. Fig. 4 shows the difference between the three FSConvs in the channel direction. The pseudocode of FSConv is given in Algorithm 1, where G is set to 3 to make it easier to understand.

Compared with the standard convolution and self-attention, FSConv has the following characteristics.

1) Standard convolution shares the same and fixed convolution kernel in all positions, as given in Fig. 3(a). FSConv generates an independent dynamic kernel for each position.
2) FSConv uses the convolution layer as the weight generation function instead of the similarity (dot-product) between feature vectors in self-attention. Experiments in Section IV-D show that FSConv decouples different resolution features and improves the precision with less computation.

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Algorithm 1 FSConv Pseudocode, PyTorch-Like

```
# Initialization: G=3, O=C
import torch
import torch.nn as nn
X=torch.cat([Xa,Xb,Xc]) #(C*G,H,W)
if FSConv V1:
   func=nn.conv2d(C*G,O*C*G*1*1,groups=1)
   kernel=func(X)
   kernel.view(O,C*G,H,W)
   X.view(1,C*G,H,W)
   output=(kernel*X).sum(1) #(O,H,W)
if FSConv V2:
   func=nn.conv2d(C*G,O*1*G*1*1,groups=1)
   kernel=func(X)
   kernel.view(G,O,H,W)
   kernel=softmax(kernel,dim=0)
   output=Xa*kernel[0]+Xb*kernel[1]+\
   Xc*kernel[2] #(O,H,W)
if FSConv V3:
   func=nn.conv2d(C*G,1*1*G*1*1,groups=G)
   kernel=func(X)
   kernel.view(G,1,H,W)
   kernel=softmax(kernel,dim=0)
   output=Xa*kernel[0]+Xb*kernel[1]+\
   Xc*kernel[2] #(O,H,W)
```

C. Analyzing 1 × 1 Convolution Layer Through Hypersphere Embedding

The cross-entropy loss combined with softmax and 1 × 1 convolution layer is one of the most commonly used supervision components in semantic segmentation methods. The 1 × 1 convolution layer is equivalent to applying $y = W^T x + b$ on each feature vector in the feature maps. According to cosine law, $y = W^T x + b$ can be rewritten as $y = \|W^T\|\|x\|\cos(\theta) + b$. Hence, the cross-entropy loss can be written as

$$L_{\mathrm{CE}} = \frac{1}{N}\sum_{n=1}^{N} -\log\left(\frac{e^{\|w_{y_n}\|\|x_n\|\cos(\theta_{y_n,n})+b_{y_n}}}{\sum_{k=1}^{K} e^{\|w_{y_k}\|\|x_n\|\cos(\theta_{y_k,n})+b_{y_k}}}\right) \quad (6)$$

where N is the number of points in the feature maps, K is the number of categories, and y_n represents the category label of x_n. We consider that W^T consists of multiple class axes w, as shown in Fig. 5. Then, the optimization objective of formula (6) is to minimize the weighted projection from x to its corresponding class axis w. If the computation is constrained to a hypersphere manifold through zeroing bias b, normalizing W and x, the optimization objective becomes to minimize the angle between x and its corresponding w, as given in Fig. 5(c) and the following formula:

$$L_{\mathrm{CE}} = \frac{1}{N}\sum_{n=1}^{N} -\log\left(\frac{e^{\cos(\theta_{y_n,n})}}{\sum_{k=1}^{K} e^{\cos(\theta_{y_k,n})}}\right). \quad (7)$$

In order to find a better form of 1 × 1 convolution layer, we introduce the hypersphere embedding theory. Specifically, we design three variants of $y = W^T x + b$ based on hypersphere embedding to explore the effect of b, W, and x, as given in Fig. 5. Then, the analysis process is divided into training and inference two parts and ten strategies, as depicted in Table II. The training part refers to training the neural network with different variants of 1 × 1 convolution layer to determine which variant can obtain a network with stronger representation ability. The inference part is to change the structure of 1 × 1 convolution layer with the fixed neural network to find the best strategy with the highest precision.

TABLE II
TEN STRATEGIES OF THE 1 × 1 CONVOLUTION LAYER

Train \ Infer	$W^T x + b$	$\|W^T\|\|x\|cos(\theta)$	$\|x\|cos(\theta)$	$cos(\theta)$
$W^T x + b$	A	B	C	D
$\|W^T\|\|x\|cos(\theta)$		E	F	G
$\|x\|cos(\theta)$			H	I
$cos(\theta)$				J

The analysis process is described in Section IV-E, and the conclusion can be summarized as follows.

1) Bias b can be viewed as a slack variable during training that relaxes the stiff condition of linear separability in some intractable positions. Removing b when inference can improve the segmentation accuracy in these intractable positions, especially the edges.
2) The existence of $\|W\|$ and $\|x\|$ can enlarge the angle between class axes to facilitate the optimization of the neural network.

IV. EXPERIMENTS

To test the ability of the FSHRNet, we first conduct qualitative and quantitative experiments by comparing with some widely used and state-of-the-art methods on the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the iSAID dataset [62]. Then, some ablation studies are conducted on the Aligned HRNet backbone and FSConv. Finally, we analyze the 1 × 1 convolution layer from the perspective of hypersphere embedding to find the best strategy for training and inference.

A. Implementation Details

1) Datasets: Experiments are conducted on the following three datasets.

a) ISPRS Vaihingen dataset: It contains 33 orthorectified image patch mosaics with three spectral bands (red, green, and near-infrared) and 33 DSMs. The spatial resolution is 9 cm, and the average size is 2494 × 2064. There are six categories have been defined in this dataset. Among previously opened ground truths, 11 images are used to train the network, five images (ID 11, 15, 28, 30, and 34) to validate, and 17 images to test. Both images and DSMs are used in experiments.

b) ISPRS Potsdam dataset: It consists of 38 high-resolution images with a spatial resolution of 5 cm and four bands (red, green, blue, and near-infrared). Meanwhile, the DSMs are also available. The average size is 6000 × 6000, and each image is annotated with six classes as the Vaihingen dataset. Among them, 17 images except for image named 7_10 with error annotations are selected for training, six images (ID 2_12, 3_12, 4_12, 5_12, 6_12, and 7_12) for validating, and 14 images for testing. Both images and DSMs are used in experiments.

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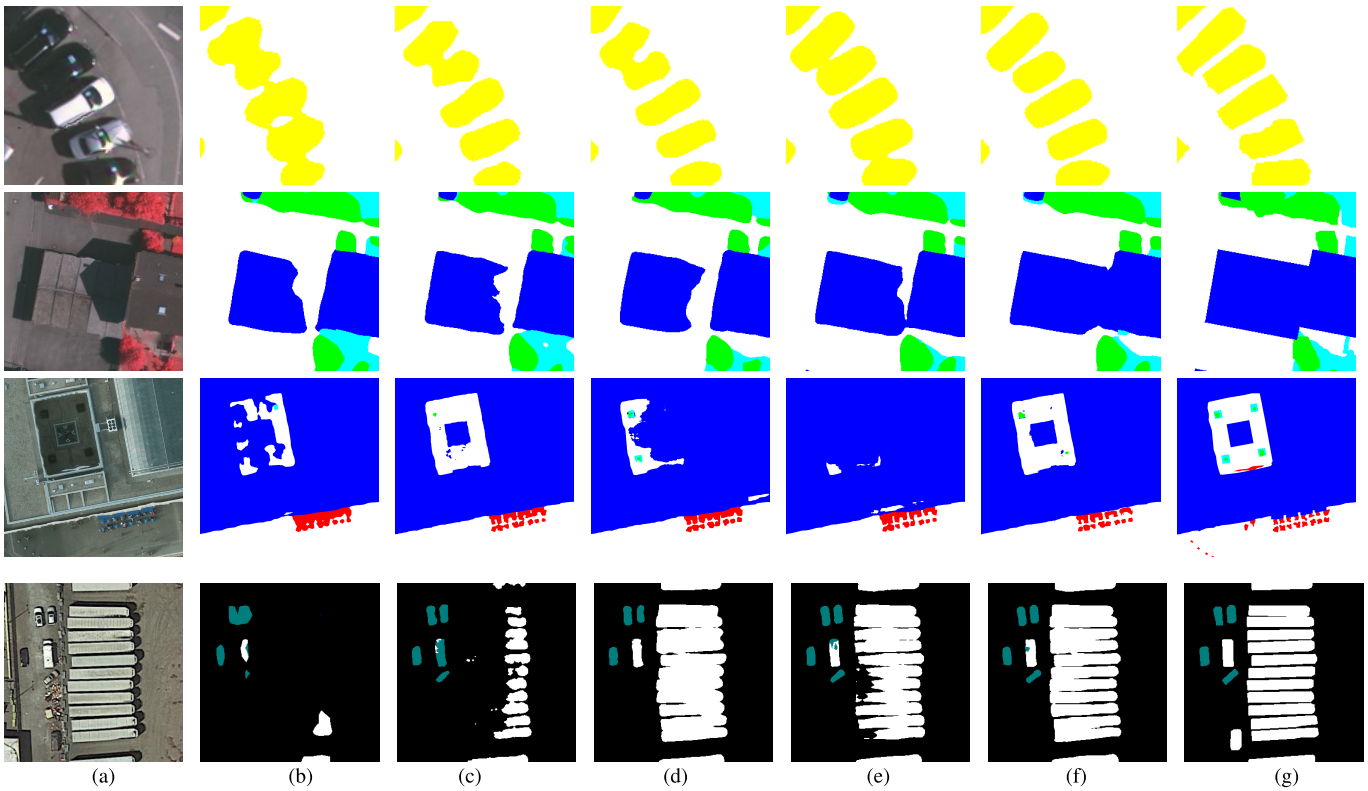


Fig. 6. Segmentation results on image patches. (a) Images. (b) DANet. (c) DeepLabV3+. (d) Swin-MLA. (e) HRNetV2. (f) FSHRNet. (g) Labels.

c) iSAID dataset [62]: iSAID consists of 2806 high-resolution remote sensing images from about 800 × 800 pixels to about 4000 × 13 000 pixels with 15 foreground classes and one background class. It is a large-scale dataset for instance segmentation, but we only use the semantic masks in this work. The images are cropped into a fixed size of 800 × 800 in advance using a sliding window striding 200 pixels. Then, 28 249 images are acquired for training, 9581 for validating, and 19 377 for testing. As the labels of the test set are unavailable, the validation set is divided into a local validation set (4000 images) and a local test set (5581 images).

For the iSAID dataset, we apply horizontal flipping and rotation for data augmentation. For the other datasets, the images are randomly cropped to 512 × 512 when training and applied random scaling (0.5 to 2.0), horizontal flipping, and rotation for data augmentation.

2) Default Network Settings: Unless otherwise specified, the Aligned HRNet backbone has four branches, and the channels from high-to-low resolution branches are 48, 96, 192, and 384. The backbone is pretrained on ImageNet. The default set of FSConv is FSConv V2 with 1 × 1 convolution layer as weight generation function. The FSHRNet is trained by strategy B in Table II.

3) Settings of Training and Inference: For the iSAID dataset, all models perform 105k iterations with four batch sizes using the AdamW optimizer where the learning rate starts from 0.0001 and decreases by a factor of 0.1 at 55k and 85k iterations. For ISPRS Vaihingen and Potsdam datasets, models are trained with 80k iterations and four batch sizes using the stochastic gradient descent (SGD) optimizer with a 0.01 initial learning rate. The learning rate is gradually decrease in the manner of $0.01 * (1 - (\text{iter}/\text{max_iter})^{0.9})$. When inference, the result is an average of three segmentation score maps with different scales (0.5, 1.0, 1.5). DSMS is introduced by directly concatenating with images.

B. Comparison Experiments

1) Qualitative Experiment: We first qualitatively compare our model with some widely used and state-of-the-art methods on the ISPRS Vaihingen, Potsdam, and iSAID datasets. Figs. 6 and 7 show some segmentation results of image patches and large-scale images, respectively, where all methods are implemented by ourselves under the same training conditions. Among them, DeepLabV3+ and DANet use ResNet-101 as the backbone. HRNetV2 and our FSHRNet are HRNet-based methods. Swin-MLA is a combination of Swin transformer [28] and multilevel feature aggregation (MLA) of SETR [51]. As depicted in the figures, the large downsampling factor in DeepLabV3+ and DANet causes the failure to accurately identify small objects, such as cars and ships, which does not appear in HRNetV2 and FSHRNet. Swin-MLA performs well on the iSAID datasets but not on the ISPRS dataset. As an extension of HRNet, our FSHRNet outputs clearer results with fewer holes and trivial patches compared with HRNetV2.

2) Quantitative Experiment: Tables III–V give the results of the quantitative experiment on the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the iSAID dataset, in which

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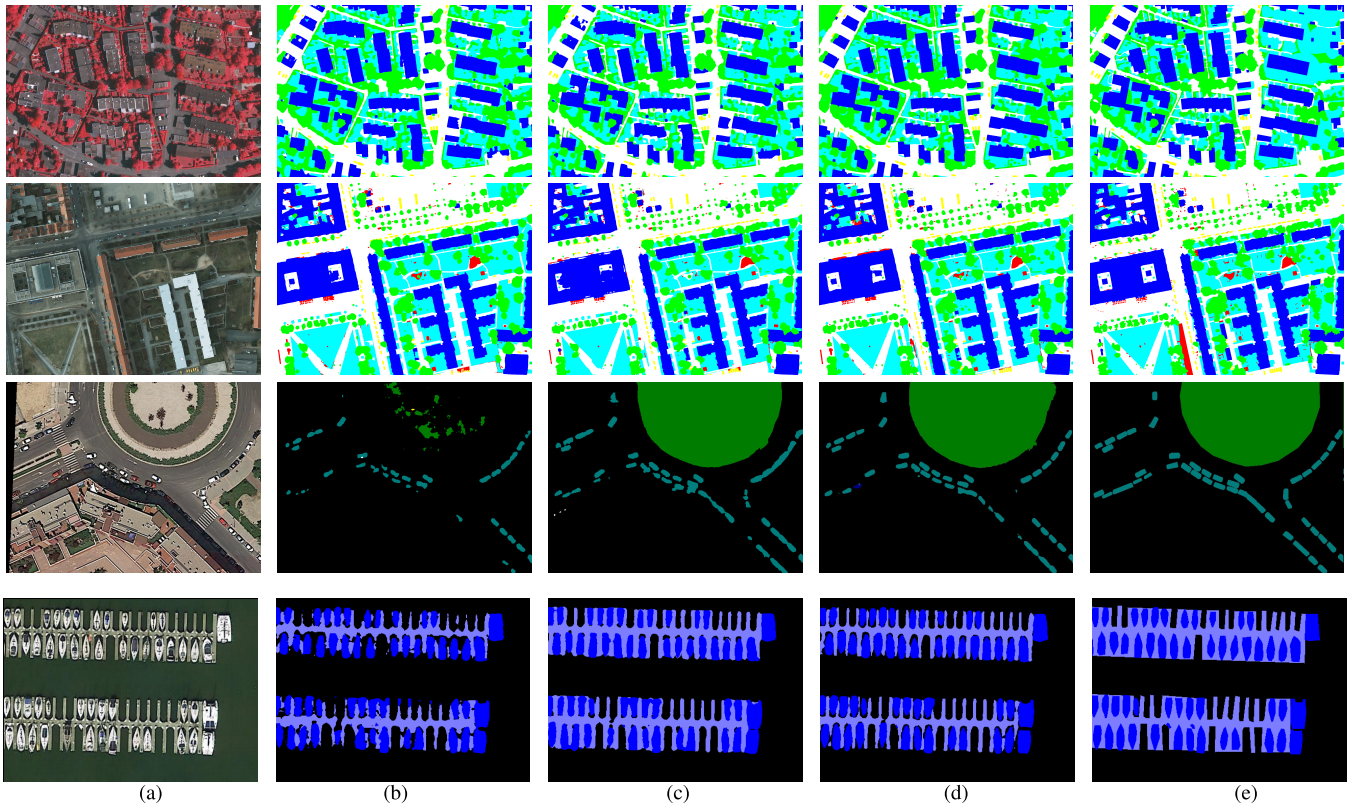


Fig. 7. Segmentation results of large-scale scenes. The three methods have different backbones. (a) Images. (b) DeepLabV3+. (c) Swin-MLE. (d) FSHRNet. (e) Label.

TABLE III

QUANTITATIVE EXPERIMENT RESULTS ON THE ISPRS VAIHINGEN DATASET. THE FSHRNET USES FSCONV V2 WITH 1 × 1 CONVOLUTION LAYER AS THE WEIGHT GENERATION FUNCTION. BOUNDARIES ARE INVOLVED IN PRECISION CALCULATION TO HIGHLIGHT GAPS BETWEEN DIFFERENT METHODS. ACCORDING TO THE ORIGINAL PAPER [54], THE RESUNET-A D6 DOES NOT USE WEIGHTS FROM PRETRAINED NETWORKS. THE BACKBONE OF ACBNET [63] IS CHANGED FROM RESNET-18 TO RESNET-101 TO ENSURE THE FAIRNESS OF THE COMPARISON. THE BACKBONE OF THE EFFICIENT MODEL [64] IS EFFICIENT MULTISCALE VISION TRANSFORMER (REST) [65]

Method	Backbone	Imp. Surf.	Build	Low Veg.	Tree	Car	Validation Set			Test Set		
							Mean F1	OA(%)	mIoU(%)	Mean F1	OA(%)	mIoU(%)
FCN-8s	VGG-16	88.46	92.06	79.53	86.03	62.25	81.39±0.04	85.31±0.03	69.66±0.05	81.66±0.02	86.39±0.02	70.26±0.03
ResUNet-a d6	ResNet-like	89.47	93.49	80.06	86.28	67.65	84.18±0.05	86.19±0.03	73.27±0.07	83.39±0.03	87.24±0.02	72.49±0.04
DANet	ResNet-101	89.72	93.55	80.94	86.63	69.41	85.02±0.05	86.93±0.03	74.50±0.07	84.05±0.02	87.63±0.01	73.36±0.02
FCN-8s	ResNet-101	89.66	93.55	80.51	86.67	73.03	85.37±0.02	86.91±0.01	74.98±0.03	84.69±0.02	87.60±0.01	74.10±0.03
Efficient-B	ResT-B	90.02	93.92	80.05	86.34	73.15	85.74±0.03	87.32±0.04	75.53±0.05	84.70±0.02	87.61±0.02	74.15±0.03
Swin+MLA	Swin-S	89.83	93.64	80.94	86.65	72.81	85.58±0.04	87.05±0.02	75.28±0.05	84.78±0.05	87.75±0.03	74.28±0.07
DeepLabV3+	ResNet-101	89.95	93.83	80.78	86.77	73.05	85.70±0.02	87.02±0.02	75.43±0.03	84.88±0.01	87.82±0.01	74.41±0.02
Simple 4× Net		89.83	93.11	80.60	86.95	75.79	86.03±0.05	87.08±0.04	75.92±0.08	85.22±0.03	87.62±0.03	74.82±0.03
PSPNet	ResNet-101	90.14	93.96	81.21	87.05	75.67	86.29±0.04	87.38±0.01	76.35±0.06	85.61±0.03	88.11±0.01	75.40±0.05
ABCNet	ResNet-101	90.23	94.11	**81.57**	86.94	75.83	86.15±0.02	87.18±0.02	76.09±0.04	85.73±0.02	88.19±0.01	75.58±0.03
HRNetV2	HRNet	90.24	94.30	80.88	86.97	79.64	86.95±0.06	87.39±0.03	77.31±0.08	86.41±0.04	88.20±0.03	76.51±0.05
FSHRNet	HRNet	**90.34**	**94.31**	81.37	**87.19**	**80.10**	**87.05±0.04**	**87.45±0.05**	**77.46±0.06**	**86.66±0.03**	**88.38±0.03**	**76.86±0.04**

the backbone is emphasized in the second column. ResNet-based PSPNet with 8 downsampling factor shows better performance than FCN-8s, DANet, and DeepLabV3+ whose downsampling factor of ResNet is 16. ABCNet [63] achieves the highest accuracy based on the ResNet by introducing a spatial path and an attention enhancement module. Simple4× Net shows comparable performance to ResNet-based methods on ISPRS datasets but cannot handle the long-tailed distributions of the iSAID dataset. The transformer-based Swin + MLA and Efficient [64] achieve the best accuracy on the iSAID dataset but perform relatively poorly on ISPRS datasets in which the objects are dense and co-occurring. The HRNetV2 and FSHRNet show good accuracy on all three datasets, and our FSHRNet performs best on ISPRS Vaihingen and Potsdam datasets.

The data in the three tables also reflect a problem: the improvements brought by FSHRNet over HRNetV2 on ISPRS Vaihingen and Potsdam datasets are quite marginal compared to the iSAID dataset. One possible reason for the above phenomenon is that the images in ISPRS datasets have a

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TABLE IV

QUANTITATIVE EXPERIMENT RESULTS ON THE ISPRS POTSDAM DATASET. THE FSHRNET USES FSCONV V2 WITH 1×1 CONVOLUTION LAYER AS THE WEIGHT GENERATION FUNCTION. BOUNDARIES ARE INVOLVED IN PRECISION CALCULATION TO HIGHLIGHT GAPS BETWEEN DIFFERENT METHODS

Method	Backbone	Imp. Surf.	Build	Low Veg.	Tree	Car	Validation Set			Test Set		
							Mean F1	OA(%)	mIoU(%)	Mean F1	OA(%)	mIoU(%)
FCN-8s	VGG-16	90.08	95.11	83.48	83.42	87.05	88.26±0.01	89.48±0.02	79.33±0.02	87.83±0.03	87.33±0.02	78.58±0.03
ResUNet-a d6	ResNet-like	90.49	95.99	83.67	83.72	86.13	88.49±0.02	89.86±0.02	79.76±0.03	88.00±0.04	87.65±0.05	78.90±0.06
DANet	ResNet-101	91.07	96.36	85.02	85.83	86.45	89.13±0.09	90.56±0.06	80.75±0.14	88.95±0.02	88.77±0.03	80.37±0.03
Simple 4× Net		90.90	96.17	85.06	84.97	88.62	89.34±0.06	90.53±0.10	81.08±0.10	89.15±0.01	88.46±0.02	80.68±0.01
FCN-8s	ResNet-101	91.43	96.37	85.33	86.21	88.33	89.73±0.06	90.90±0.06	81.68±0.09	89.54±0.02	89.09±0.01	81.30±0.02
Swin+MLA	Swin-S	91.36	96.30	85.10	85.00	89.66	90.33±0.02	91.25±0.03	82.65±0.03	89.49±0.03	88.81±0.04	81.24±0.05
Efficient-B	ResT-B	91.19	96.29	85.50	86.05	90.12	90.14±0.04	91.13±0.09	82.39±0.07	89.83±0.01	89.03±0.03	81.77±0.03
PSPNet	ResNet-101	91.58	96.46	85.57	86.31	89.37	90.20±0.09	91.28±0.08	82.45±0.05	89.86±0.02	89.30±0.04	81.82±0.04
DeepLabV3+	ResNet-101	91.70	96.45	85.48	86.37	89.83	90.19±0.01	91.24±0.01	82.44±0.01	89.97±0.06	89.32±0.05	82.00±0.08
ABCNet	ResNet-101	91.72	96.50	85.59	86.81	89.30	90.37±0.03	91.21±0.02	82.68±0.03	89.98±0.05	89.36±0.03	82.07±0.05
HRNetV2	HRNet	92.00	96.57	86.04	86.78	91.36	90.67±0.04	91.36±0.04	83.24±0.06	90.55±0.05	89.65±0.04	82.96±0.06
FSHRNet	HRNet	**92.16**	**96.59**	**86.12**	**86.94**	**91.53**	**90.79±0.03**	**91.45±0.02**	**83.43±0.05**	**90.67±0.04**	**89.82±0.03**	**83.16±0.06**

TABLE V

QUANTITATIVE EXPERIMENT RESULTS ON THE ISAID DATASET. THE 15 CATEGORIES ARE SHIP (SH), STORAGE TANK (ST), BASEBALL DIAMOND (BD), TENNIS COURT (TC), BASKETBALL COURT (BC), GROUND TRACK FIELD (GTF), BRIDGE (BR), LARGE VEHICLE (LV), SMALL VEHICLE (SV), HELICOPTER (HE), SWIMMING POOL (SP), ROUNDABOUT (RO), SOCCER BALL FIELD (SBF), PLANE (PL), AND HARBOR (HA)

Method	backbone	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HE	SP	RO	SBF	PL	HA	mF1(Val)	mF1(Test)
Simple 4× Net		62.50	54.97	34.86	85.33	36.87	29.69	5.76	64.27	54.05	0.13	43.19	3.43	56.40	70.45	41.92	46.37±0.84	42.92±0.59
DeepLabV3+	ResNet-101	65.10	61.62	62.73	89.72	71.70	59.96	27.10	65.42	54.75	18.21	44.17	50.46	71.72	76.21	48.73	59.37±0.11	57.84±0.59
FCN-8s	ResNet-101	62.96	63.12	61.76	90.35	71.56	59.33	30.91	65.08	47.97	21.76	44.20	49.84	77.15	75.90	45.80	60.31±0.57	57.85±0.41
DANet	ResNet-101	63.37	64.24	69.44	90.37	72.17	60.06	30.65	65.67	47.29	27.82	50.66	58.56	75.27	76.05	46.16	61.68±1.07	59.85±0.38
FCN-8s	VGG-16	70.06	67.28	73.21	93.11	69.51	60.59	34.99	70.49	58.88	28.54	59.09	76.17	71.21	84.38	55.37	59.87±0.37	64.86±0.16
PSPNet	ResNet-101	70.62	65.74	72.22	92.99	77.05	61.25	44.03	70.21	54.85	28.31	51.23	77.15	77.79	80.92	55.92	65.03±0.29	65.35±0.26
ABCNet	ResNet-101	74.43	64.87	85.36	94.80	73.64	62.99	46.63	75.11	60.81	8.78	64.32	80.74	64.29	67.51	61.38	63.23±0.73	65.72±0.47
HRNetV2	HRNet	71.14	70.16	74.77	92.13	76.07	60.74	45.86	73.00	62.91	39.31	57.60	64.71	78.89	82.89	53.95	67.53±0.20	66.94±0.17
FSHRNet	HRNet	71.31	69.39	76.64	91.39	**78.74**	62.95	**47.51**	72.85	62.92	33.18	59.84	69.29	**79.12**	83.22	54.44	**68.48±0.37**	67.52±0.20
Swin+MLA	Swin-S	78.55	71.62	**82.84**	**94.99**	76.81	71.77	38.01	74.63	61.30	**42.45**	61.96	**87.48**	77.07	88.72	63.39	67.89±0.58	71.44±0.14
Efficient-B	ResT-B	**80.81**	**72.13**	77.69	94.95	74.05	**71.92**	42.51	**78.48**	**67.62**	37.76	**67.64**	79.28	76.91	**89.19**	**65.73**	66.05±0.20	**71.78±0.16**

small ground sampling distance (GSD) than the images in the iSAID dataset. Hence, we conduct an experiment to verify whether GSD has an effect on accuracy by downsampling the images of the ISPRS Vahingen dataset and recording the experimental results in Table VI. Table VI provides that the mean F1 and mIoU of FSHRNet show a significant improvement with the increase in GSD, but OA does not improve compared to HRNetV2. Then, we plot Fig. 8 to analyze which regions are correctly classified in FSHRNet compared to HRNetV2. A qualitative conclusion can be drawn from Fig. 8 that FSHRNet has better segmentation results on small-scale objects (cars) and boundaries between objects. When the GSD becomes larger, the size of some objects in images becomes smaller, which makes FSHRNet perform better on mean F1 and mIoU. Dense small-scale objects are also widely distributed in the iSAID dataset so that the advantage of FSHRNet can be reflected. However, due to the insensitivity of overall accuracy (OA) on small-scale objects, the advantage of FSHRNet cannot be reflected, resulting in an insignificant improvement in OA.

TABLE VI

IMPROVEMENT OF FSHRNET V2 OVER HRNETV2 UNDER DIFFERENT GSDS ON THE ISPRS VAHINGEN DATASET

GSD	Method	Val			Test		
		mean F1	OA(%)	mIoU(%)	mean F1	OA(%)	mIoU(%)
9cm	HRNetV2	86.95	87.39	77.31	86.41	88.20	76.51
	FSHRNet	**87.05**	**87.45**	**77.46**	**86.66**	**88.38**	**76.86**
18cm	HRNetV2	85.49	86.74	75.12	84.10	87.08	73.28
	FSHRNet	**85.70**	**86.96**	**75.45**	**84.54**	**87.23**	**73.85**
27cm	HRNetV2	84.50	86.37	73.70	83.41	86.59	72.32
	FSHRNet	**84.92**	**86.49**	**74.13**	**83.82**	**86.77**	**72.84**

C. Ablation Study for High-Resolution Backbone

1) Number of Branches: Default Aligned HRNet backbone extends three low-resolution branches to enhance the performance of high-resolution features. To test the effect of the added low-resolution branches, we conduct an ablation study by changing the branch numbers of the FSHRNet. The experiment result provided in Table VII provides that adding low-resolution branches can comprehensively improve the segmentation accuracy. Meanwhile, benefiting from the large receptive field of low-resolution features, the multibranch structure shows superiority when dealing with large objects, as illustrated in Fig. 9. Furthermore, the decline of accuracy does not appear on small objects (e.g., cars) with the addition of low-resolution features.

2) Efficiency of High-Resolution Backbone: Table VIII provides the efficiency comparison results with some widely used backbones and segmentation methods. Since the channels of low-branch increase exponentially, the GFLOPs and parameters of HRNet increase significantly with the addition of branches. Large inference memory occupancy also occurs in HRNet due to the high-resolution features. Compared with

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TABLE VII

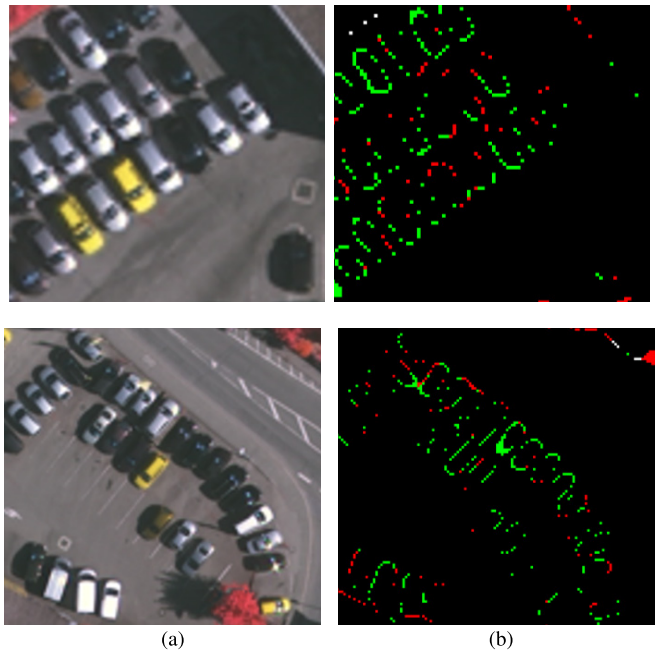
ABLATION STUDY ON BRANCH NUMBERS OF FSHRNET ON THE ISPRS VAIHINGEN TEST SET

Num of Branches	Imp. Surf.	Build	Low Veg.	Tree	Car	Mean F1	OA(%)	mIoU(%)
1	89.77	93.07	80.54	86.95	75.79	85.22±0.03	87.58±0.02	74.77±0.04
2	90.18	93.78	81.23	**87.29**	79.88	86.47±0.03	88.16±0.02	76.55±0.04
3	90.25	94.14	81.20	87.13	79.81	86.51±0.02	88.25±0.01	76.62±0.02
4	**90.34**	**94.31**	**81.37**	87.19	**80.10**	**86.66±0.03**	**88.38±0.03**	**76.86±0.04**

TABLE VIII

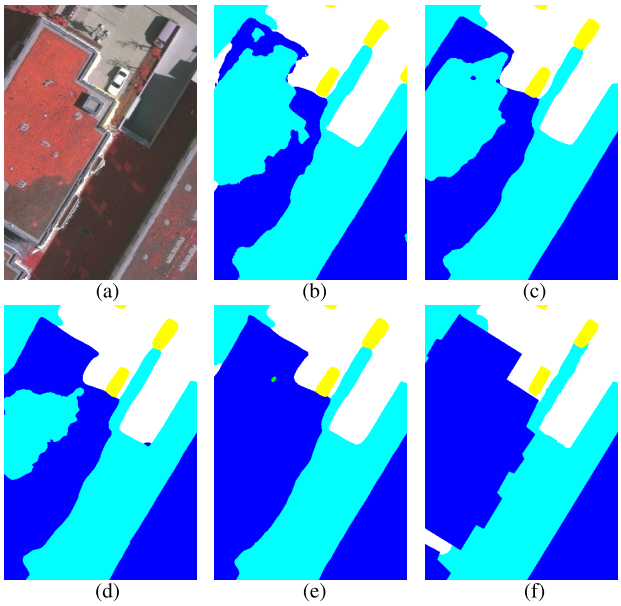
EFFICIENCY COMPARISON OF THE HIGH-RESOLUTION BACKBONE WITH SOME WIDELY USED BACKBONES AND SEGMENTATION MODELS IN TERMS OF GFLOPS, PARAMETERS, AND INFERENCE MEMORY. THE THREE INDICATORS ARE DIVIDED INTO TWO COLUMNS TO HIGHLIGHT THE EFFICIENCY OF BACKBONE AND COMPLETE MODEL, RESPECTIVELY. HRNET-N MEANS THAT THE NUMBER OF BRANCHES IS N. THE BATCH SIZE IS SET TO 1 WHEN CALCULATING GFLOPS, PARAMETERS, AND INFERENCE MEMORY WITH 512 × 512 IMAGE SIZE. THE FSHRNET USES FSCONV V3 WITH 1 × 1 CONVOLUTION LAYER AS THE WEIGHT GENERATION FUNCTION

Method	Backbone	Mean F1	OA(%)	GFLOPs		#Params(MB)		Infer Memory(MB)	
				Backbone	Method	Backbone	Method	Backbone	Method
FCN-8s	VGG-16	81.67	86.38	80.50	80.66	14.72	15.31	**361.50**	**368.26**
DeepLabV3+	ResNet-101(16×)	84.86	87.83	52.27	89.04	42.50	59.34	686.20	836.47
PSPNet	ResNet-101(8×)	85.61	88.12	178.08	265.13	42.50	67.94	1760.20	1872.80
Swin-MLA	Swin-S	84.87	87.79	47.90	49.64	48.84	51.00	1034.91	1103.41
FSHRNet	Aligned HRNet-1	85.22	87.58	**29.19**	**29.20**	**1.77**	**1.78**	639.08	645.08
FSHRNet	Aligned HRNet-2	86.47	88.16	53.61	53.63	7.69	7.70	948.66	962.91
FSHRNet	Aligned HRNet-3	86.51	88.25	75.45	75.47	28.57	28.58	1081.09	1095.46
HRNetV2	HRNet-4	86.49	88.22	85.17	93.78	65.33	65.85	1053.55	1237.74
FSHRNet	Aligned HRNet-4	**86.57**	**88.32**	85.37	85.39	65.42	65.44	1130.80	1145.30



(a) (b)

Fig. 8. Visualization results of which regions are correctly classified in FSHRNet V2 compared to HRNetV2 on the ISPRS Vaihingen dataset with 27-cm GSD. Green points represent the correct changes (misclassified in HRNetV2 but correctly classified in FSHRNet), and red points mean incorrect changes (correctly classified in HRNetV2 but misclassified in FSHRNet). (a) Images. (b) Visualization results.



(a) (b) (c) (d) (e) (f)

Fig. 9. Segmentation results of large object with different numbers of branches. (a) Image. (b) Result of one branch. (c) Result of two branches. (d) Result of three branches. (e) Result of four branches. (f) Label.

VGG, ResNet, and Swin transformer, the efficiency of the multibranch HRNet backbone has no advantages. Instead, our FSHRNet only adds a few parameters and calculations to the HRNet backbone. If accuracy is considered, the advantage of high-resolution features is obvious; for instance, the one-branch FSHRNet can achieve accuracy close to DeepLabV3+ with very few parameters. In addition, the FSHRNet has higher precision and efficiency compared to HRNetV2.

D. Ablation Study for FSConv Layer

1) Comparison With Other Context Aggregation Modules: FSConv, including three variants, is proposed to fuse multiresolution features. To test its effect fairly, we use the Aligned HRNet with concatenate operation as a baseline network and compare the performance of different context aggregation

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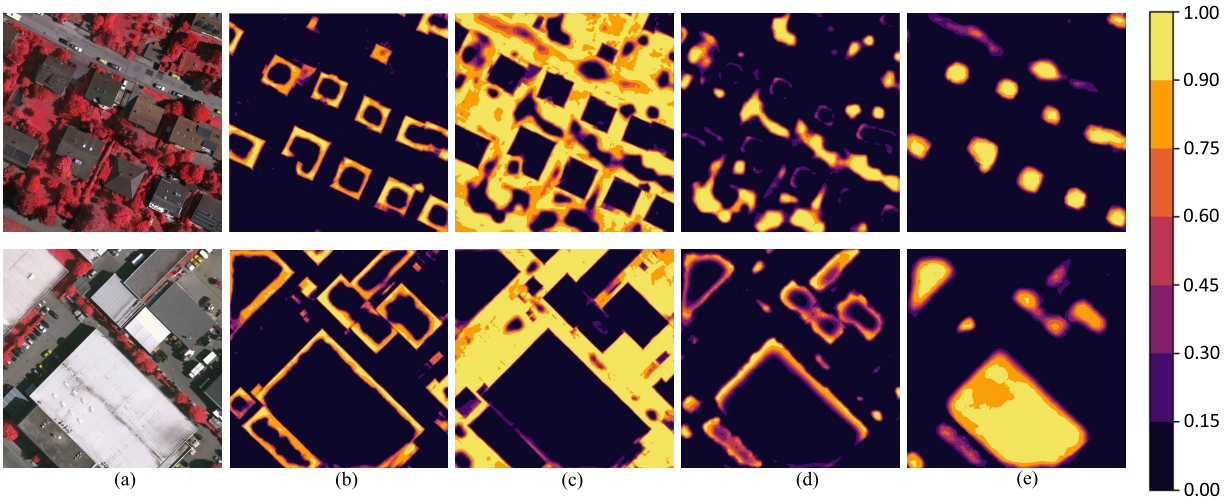


Fig. 10. Visualization results of FSConv V3 (1 × 1) weight maps. (a) Images. (b) Weight map of branch 1. (c) Weight map of branch 2. (d) Weight map of branch 3. (e) Weight map of branch 4.

TABLE IX
COMPARISON RESULTS OF FSCONV WITH SOME CONTEXT AGGREGATION APPROACHES ON THE ISPRS VAIHINGEN DATASET. ALL MODULES ARE IMPLEMENTED BY OURSELVES. THE BASELINE NETWORK AND FSCONV USE A SINGLE 1 × 1 CONVOLUTION LAYER TO GENERATE THE SEGMENTATION RESULTS. – MEANS OUT OF MEMORY. ($k \times k$) AFTER FSCONV MEANS THE KERNEL SIZE OF THE WEIGHT GENERATION FUNCTION. REFERENCES: OCR [10] AND SK CONV [61]

Method	Mean F1	OA (%)	mIoU (%)	Infer Memory(MB)
Aligned HRNet (baseline)	86.44±0.02	88.08±0.02	76.51±0.03	
+ASPP	86.44±0.01	88.24±0.01	76.53±0.01	318.01
+PPM	86.43±0.03	88.34±0.01	76.54±0.04	166.15
+SK Conv	86.35±0.02	88.28±0.02	76.42±0.02	14.01
+Dual Attention	-	-	-	-
+OCR	85.65±0.04	87.99±0.03	75.43±0.04	229.92
+FSConv V1 (1×1)	-	-	-	-
+FSConv V2 (1×1)	86.66±0.03	88.38±0.03	76.86±0.04	78.00
+FSConv V2 (3×3)	86.69±0.02	88.30±0.01	76.90±0.03	78.00
+FSConv V3 (1×1)	86.57±0.02	88.32±0.02	76.74±0.04	14.50
+FSConv V3 (3×3)	86.55±0.01	88.38±0.02	76.70±0.03	14.50

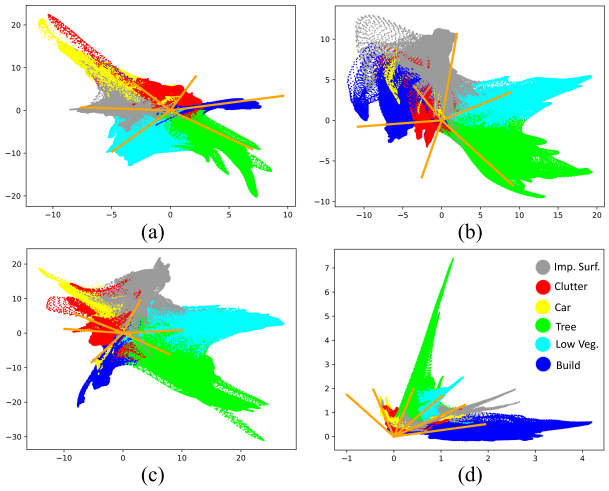


Fig. 11. Visualization of class axes and feature points on the ISPRS Vaihingen dataset. The formulas below subfigures represent which variant of the 1 × 1 convolution layer is used to train the network. The dimensionality reduction is implemented by adding an extra convolutional layer before the last convolutional layer and then directly visualizing the 2-D feature points. (a) $W^T x + b$. (b) $\|W^T\|\|x\|\cos(\theta)$. (c) $\|x\|\cos(\theta)$. (d) $\cos(\theta)$.

approaches by adding them to the baseline network. Note that the baseline network and the FSConv module use a single 1 × 1 convolution layer to produce the segmentation result; the other modules follow their original structure. Comparison results are illustrated in Table IX. Taken together, the following conclusions can be summarized: 1) ASPP, PPM, and OCR occupy large memory but bring low gain; b) compute-intensive modules, such as dual-attention and FSConv V1, may lead to memory overflow when used in high-resolution features; and 3) compared with FSConv V3, FSConv V2 achieves better accuracy but consumes more memory. Comparatively speaking, our FSConv is a relatively better choice to enhance the representation of multiresolution features.

2) Visualization of FSConv: The goal of the FSConv layer is to fuse the multiresolution features by dynamic weights so that features with different resolutions can focus on different information. Intuitively, small objects and edges can be well identified by high-resolution features so that the high-resolution branch should be assigned more weight to recognize them. On the other hand, a low-resolution branch with a large receptive field should contribute to the integrity of large objects. We verify this assumption by visualizing the weights of four resolution branches in FSConv V3 (1 × 1). The visualization results are given in Fig. 10, which basically meets the previous assumptions. Weight maps of branches 1 and 2 (high-resolution) pay more attention to the edges and small objects, such as cars. Branch 4 focuses on the internal structure of the objects to ensure the integrity of segmentation results.

E. Discussion of 1 × 1 Convolution Layer

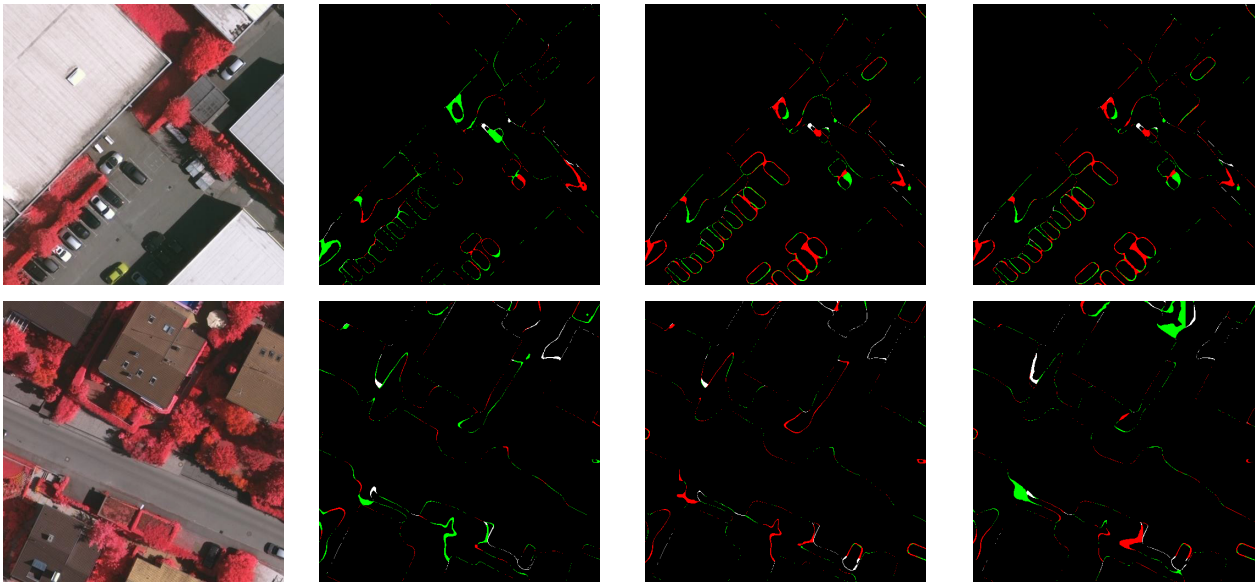
The 1 × 1 convolution layer ($y = W^T x + b$) is divided into four variants and ten strategies through the hypersphere embedding theory, as given in Table II. The purpose of

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TABLE X

SEGMENTATION ACCURACY OF DIFFERENT VARIANTS OF THE 1 × 1 CONVOLUTION LAYER ON THE ISPRS VAIHINGEN DATASET. THE NETWORK ARCHITECTURE IS FSHRNET-4 WITH FSCONV V2 (1 × 1). THE THREE ACCURACY INDICATORS ARE MEAN F1 SCORE, OA, AND MIOU IN ORDER

Train \ Infer	$W^T x + b$	$\|W^T\| \|x\| cos(\theta)$	$\|x\| cos(\theta)$	$cos(\theta)$
$W^T x + b$	A:86.48, 88.32, 76.59	**B:86.66, 88.38, 76.86**	C:86.33, 88.28, 76.39	D:86.43, 88.27, 77.53
$\|W^T\| \|x\| cos(\theta)$		E:86.49, 88.24, 76.61	F:86.56, 88.23, 76.71	G:86.54, 88.21, 76.67
$\|x\| cos(\theta)$			H:86.46, 88.29, 76.58	I:86.16, 88.25, 76.15
$cos(\theta)$				J:79.44, 87.76, 68.97



(a) (b) (c) (d)

Fig. 12. Visualization results about where the categories have changed after removing bias b, normalizing W, and normalizing x. Taking (b) as a sample to describe the process of generating this visualization result. First, training the network with $y = W^T x + b$ and outputting the segmentation result A. Then, removing the b when inference and outputting the segmentation result B. Next, using B-A to calculate the positions where the category changes after removing the b. Finally, using green points represent the correct changes (misclassified in A but correctly classified in B), red points mean incorrect changes (correctly classified in A but misclassified in B), and white points represent the invalid changes (misclassified in A and B). If there are more green points than red points, the segmentation accuracy will be improved. (a) Images. (b) Training: $W^T x + b$ Inference: $\|W^T\| \|x\| \cos(\theta)$. (c) Training: $W^T x + b$ Inference: $\|x\| \cos(\theta)$. (d) Training: $W^T x + b$ Inference: $\cos(\theta)$.

TABLE XI

EFFECT OF REMOVING BIAS UNDER DIFFERENT DATASETS AND DIFFERENT LOSS FUNCTIONS. THE NETWORK ARCHITECTURE IS FSHRNET-4 WITH FSCONV V2 (1 × 1). RB MEANS THAT REMOVING BIAS IS USED WHEN INFERENCE. REFERENCES: FOCAL IS FOCAL LOSS [66], AND LOVASZ IS THE LOVASZ-SOFTMAX LOSS [67]

Dataset	Method	Mean F1	OA(%)	mIoU(%)
Vaihingen	Cross Entropy	86.51	88.32	76.67
	Cross Entropy+RB	**86.66**	**88.38**	**76.86**
	Focal	84.68	87.68	74.13
	Focal+RB	**85.36**	**87.75**	**75.01**
	Lovasz	86.68	87.89	76.81
	Lovasz+RB	**86.79**	**87.92**	**76.98**
Potsdam	Cross Entropy	90.53	89.75	82.98
	Cross Entropy+RB	**90.67**	**89.82**	**83.16**
iSAID	Cross Entropy	-	-	67.37
	Cross Entropy+RB	-	-	**67.52**

this section is to analyze the impact of different variants of the 1 × 1 convolution layer on accuracy from the perspectives of training and inference and to find the best strategy. We conduct an experiment on the ISPRS Vaihingen dataset based on Table II and record the accuracy in Table X.

1) Analysis of Training: The training part refers to training the segmentation model with different variants of the 1 × 1 convolution layer to determine which variant can obtain a network with stronger representation ability. Combined with Section III-C, a strong representation ability should be reflected in two aspects: the angle between class axes is large, and the feature points are clustered on their corresponding class axes, that is, to minimize the intraclass distance and enlarge the interclass distance. Hence, we plot the class axes and feature points of different variants in Fig. 11 using dimensionality reduction. The angle between class axes in Fig. 11(a)–(c) is large enough, which brings a higher accuracy, as given in A, E, and H in Table X. Conversely, the small angle of Fig. 11(d) leads to low accuracy, as given in J in Table X. The misclassification of clutters caused by insufficient training data is also reflected in Fig. 11. Finally, a rough conclusion can be drawn from the analysis of the training part, that is, the existence of $\|W\|$ and $\|x\|$ can enlarge the angle between class axes to facilitate the optimization of the neural network.

2) Analysis of Inference: This part is to find the best strategy with the highest precision by changing the structure of the 1 × 1 convolution layer with the fixed trained neural network. Strategy B in Table X shows that the highest accuracy can be obtained by training the model using $y = W^T x + b$ and removing bias b during inference. Then, we draw Fig. 12 to explore which regions are correctly classified after the removal of bias. Fig. 12(b) illustrates that the changes mainly occur on

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the boundaries of objects where more pixels are classified to the correct category by comparing the green and red points. Conversely, Fig. 12(c) and (d) reveals that normalizing W and x during inference results in a large number of incorrect class changes. Finally, we conduct some supplementary experiments in Table XI to verify the generalization ability to remove bias. In a sense, bias b can be viewed as a slack variable that relaxes the stiff condition of linear separability in some intractable pixels, such as boundaries. The comparison of strategies B and E proves that the accuracy becomes worse without b as a slack variable.

V. Conclusion

We proposed an FSHRNet to realize multiscale object segmentation in VHR images based on the strong linear separability of high-resolution features. The proposal of FSConv and the improvement of 1×1 convolution layer is a tradeoff between efficiency and precision, which gives high-resolution features stronger representational ability and higher segmentation accuracy with almost no loss of efficiency. Experiments demonstrate that our model outperforms the most widely used methods on the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the iSAID dataset.

Even if the segmentation accuracy is being continuously improved, the current segmentation methods still face a problem: segmentation result often violates vision common sense, such as the irregular edges of a rectangle building and holes in an object. We think that these drawbacks are caused by the pixel-based classification paradigm. Intuitively, objects should be viewed as a whole rather than some pixels. Recently, some methods have been proposed to deal with the problems caused by the pixel-based classification paradigm, such as mask-based methods [68] and kernel-based methods [69]. This is also our future research direction.

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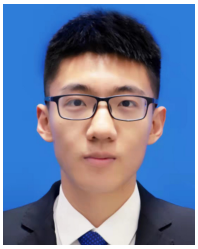
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